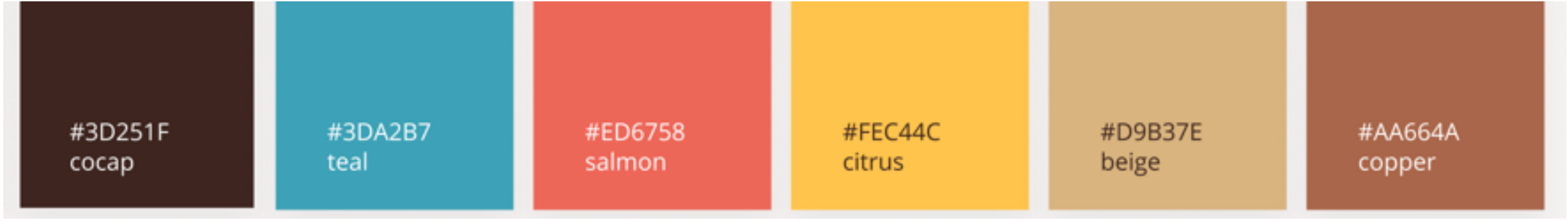




Takeaway

- ε controls the bias–variance–stability trade-off:
 - small $\varepsilon \rightarrow$ accurate but unstable/slow.
 - large $\varepsilon \rightarrow$ stable/fast but approximate.



#3D251F
cocap

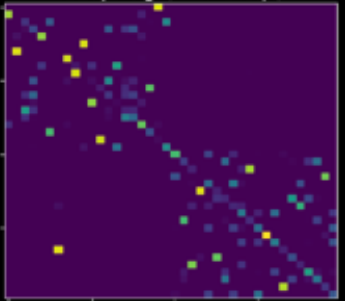
#3DA2B7
teal

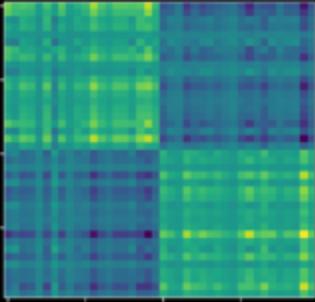
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salmon

#FEC44C
citrus

#D9B37E
beige

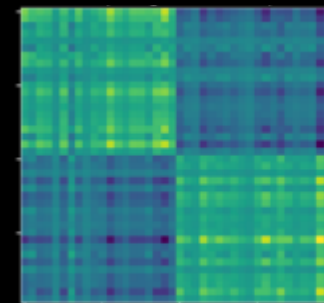
#AA664A
copper





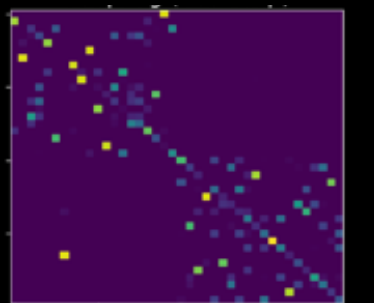
Trade-offs in Choosing ε

Large ε :



- Coupling is smoother (“blurrier” transport).
- Converges fast, numerically stable.
- But solution may be too far from true OT.

Small ε :



- Coupling π_ε close to *true* OT solution.
- Slow convergence, more iterations needed.
- Numerical issues: under/overflow in $\mathbf{K} = e^{-\mathbf{C}/\varepsilon}$.

Practical Guidelines

- **Heuristic:** set ε proportional to a typical transport cost (e.g., median of entries of $\mathbf{C}$).
- In practice (especially ML), ε often treated as a tunable hyperparameter.
- Often chosen via cross-validation.
- To simplify choice, normalize distances, e.g. $\mathbf{C} \leftarrow \mathbf{C}/\text{mean}(\mathbf{C})$, sweep $\varepsilon \in \{10^{-3}, \dots, 10^1\}$

Takeaway

- ε controls the bias–variance–stability trade-off:
 - small $\varepsilon \rightarrow$ accurate but unstable/slow.
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Barycentric Projection